

Yuanfan Chen

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EDUCATION

Cornell Tech, Cornell University

Master of Engineering in Computer Science (GPA: 3.9)

New York, NY

Aug. 2025 – May. 2026

University of Toronto

Honours Bachelor of Science, Computer Science Specialist (GPA: 3.9)

Toronto, Canada

Sep. 2020 – May. 2024

OPEN SOURCE

- **LMCache**: KV-cache hit observability for the multi-process lookup path; revived and landed L1/L2 hit attribution and early-exit reasons on the lookup event (PR #4734), then added the Prometheus counters that aggregate them per model (PR #4962); both merged.
- **vLLM**: added `gelu_tanh` to the AITER fp8 fused-MoE backend (Gemma-4-style models) and fixed a silent correctness bug where padded fp8 expert weights were allocated uninitialized, leaking pad rows into the quant scale and layer output (PR #55251, approved, in review).
- **AITER (ROCM)**: proposed the gfx942 unified-attention prefill config fix for Gemma-4 head-512 layers (PR #5649); the maintainers' dtype-split follow-up (PR #5650) benchmarks against it; contributed its end-to-end MI325X vLLM serving validation (in review).
- **SGLang-Omni**: wrote the CUDA IPC weight export/import library and MPS-DP launcher that let data-parallel replicas on one GPU share model weights (PR #1124, merged); co-developed Restage, which ranks multi-stage residency shapes from a one-GPU calibration instead of a placement grid, integration in review (PR #2134).

EXPERIENCE

Machine Learning Infra Engineer

Character.AI: *Inference Optimization*

Aug 2026 – Present

Bay Area, CA

- Own production inference optimization for large-scale **multi-turn, long-context** chat on a **vLLM + LMCache** stack serving the **Gemma, DeepSeek, and GLM** model families across **thousands of GPUs**, including latest-generation **AMD Instinct** clusters; work spans **KV-cache offloading** and **kernel optimization**.
- Deployed a **hierarchical, distributed KV cache** (GPU → L1 → shared remote tiers) with **LMCache**, raising the **prefix-cache hit rate from 70% to 90%** and eliminating most long-context prefill recompute.
- Drive kernel work from measurement: profile compute- vs. memory-bound phases and per-layer time (attention vs. FFN) against **MFU** ceilings; sustained **~3K prefill tokens/s per MI325X** on a 30B-class Gemma model.
- Built a **real-traffic replay benchmark** to validate every optimization on production workload patterns before rollout.

Machine Learning System Researcher – S4AI Lab

Cornell University

Sep 2025 – Present

New York, NY

- Engineered a scheduling framework integrating **RCSPS modeling, EDF, and Greedy** to maximize throughput under **multi-class SLO constraints**; derived **stepwise performance functions** via GPU profiling and validated performance by reproducing SOTA baselines (**Slo-serve, Sarathi**).
- Leveraged **γ -Boost Queuing Theory** to analyze **light-tailed workloads**, optimizing resource scheduling for prefill versus decode phases; achieved a **300% tail latency improvement** while maintaining Avg E2E and TTFT parity with **SJF (Shortest Job First)**.
- Developed a **Vidur-based simulator** for Multiclass SLO evaluation, conducting a class-level comparative analysis of **Colocation vs. Separation** global scheduling strategies and reproducing **PolyServe** as a baseline.

Machine Learning Engineer

Tencent: *Distributed Training and Inference*

May 2025 – Sep 2025

Beijing, CN

- Mastered parallelization strategies including **Tensor Parallelism (TP)** and **Data Parallelism (DP)**, and optimized inference engines using **PagedAttention** and **Continuous Batching**.
- Optimized **GQA** and **DSA** kernels on **NVIDIA H200 and B200** GPUs to accelerate LLM inference; profiled memory hierarchies and warp stalls using **Nsight Compute (NCU)** to alleviate bandwidth bottlenecks and maximize hardware utilization.
- Reduced pipeline bubbles by implementing **Chunk Prefill** and batched decode strategies, significantly improving throughput for long-context inference workloads.

Undergraduate Student Researcher – Far Data Lab

University of Toronto

Jan 2024 – Feb 2025

Toronto, CA

- Co-designed **dpBento** to benchmark compute/memory/network/storage of heterogeneous DPUs for data processing offload.
- Studied **RocksDB** internals; explored DPU offload for decompression/compaction to evaluate compute-storage disaggregation.

PUBLICATIONS

- **Beyond Prediction: Tail-Aware Scheduling for LLM Inference**. Yueying Li, **Yuanfan Chen**, Jiayang Chen, Esha Choukse, Haoran Qiu, G. Edward Suh, Rodrigo Fonseca, Ziv Scully, Udit Gupta. **ICML 2026**. arXiv:2606.18431.
- **Making Sense of DPU Performance for Cloud Data Processing**. Jiasheng Hu, Chihan Cui, **Yuanfan Chen**, Philip A. Bernstein, Jialin Li, Qizhen Zhang. **SoCC 2026** (17th ACM Symposium on Cloud Computing). arXiv:2504.05536.

ACADEMIC SERVICE

- **Invited Reviewer**: IEEE Journal on Selected Areas in Information Theory (JSAIT) – Special Issue on Energy and Data Efficiency in AI (2026). Reviewed submissions focusing on cutting-edge LLM inference systems.